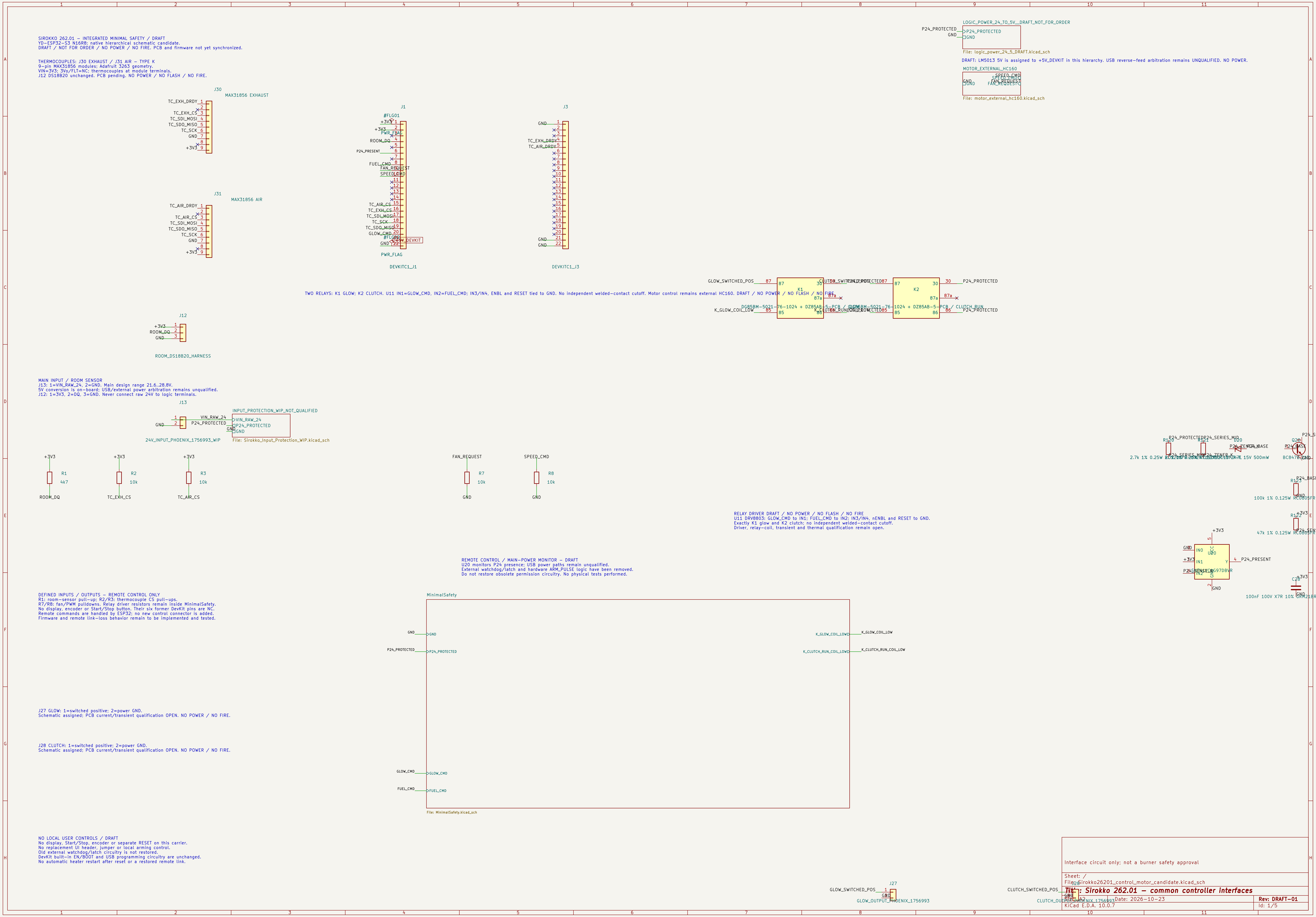

J30
MAX31856 EXHAUST
TC_EXH_DRDY
TC_EXH_CS
TC_SDI_MOSI
TC_SDO_MISO
TC_SCK
GND
+3V3
J31
MAX31856 AIR
TC_AIR_DRDY
TC_AIR_CS
J1
#FLG01
PWR_FLAG
ROOM_DQ
P24_PRESENT
FUEL_CMD
FAN_REQUEST
SPEED_CMD
GLOW_CMD
+5V_DEVKIT
DEVKITC1_J1
J3
DEVKITC1_J3
J12
ROOM_DS18B20_HARNESS
J13
VIN_RAW_24
P24_PROTECTED
24V_INPUT_PHOENIX_1756993_WIP
INPUT_PROTECTION_WIP_NOT_QUALIFIED
File: Sirokko_Input_Protection_WIP.kicad_sch
R1 4k7
R2 10k
R3 10k
R7 10k
R8 10k
LOGIC_POWER_24_TO_5V__DRAFT_NOT_FOR_ORDER
File: logic_power_24_5_DRAFT.kicad_sch
MOTOR_EXTERNAL_HC160
File: motor_external_hc160.kicad_sch
K1
K2
GLOW_SWITCHED_POS
CLUTCH_SWITCHED_POS
K_GLOW_COIL_LOW
K_CLUTCH_RUN_COIL_LOW
DG85BM-5021-76-1024 + DZ85AB-5-PCB / GLOW
DG85BM-5021-76-1024 + DZ85AB-5-PCB / CLUTCH_RUN
R120 2.7k 1% 0.25W
R121
D20 15V 500mW
Q20 BC847B,215
P24_SERIES_MID
P24_ZENER_K
P24_BASE
P24_SENSE_N
R123 100k 1% 0.125W RC0805FR-07100KL
R122 47k 1% 0.125W RC0805FR-0747KL
U20 SN74LVC1G97DBVR
IN0
IN1
IN2
VCC
Y
P24_PRESENT
C28 100nF 100V X7R 10% GRM21BR72A104K
MinimalSafety
File: MinimalSafety.kicad_sch
J27
GLOW_OUTPUT_PHOENIX_1756993
J28
CLUTCH_OUTPUT_PHOENIX_1756993

SIROKKO 262.01 – INTEGRATED MINIMAL SAFETY / DRAFT
YD-ESP32-S3 N16R8; native hierarchical schematic candidate.
DRAFT / NOT FOR ORDER / NO POWER / NO FIRE. PCB and firmware not yet synchronized.

THERMOCOUPLES: J30 EXHAUST / J31 AIR – TYPE K
9-pin MAX31856 modules; Adafruit 3263 geometry.
VIN=3V3; 3Vo/FLT=NC; thermocouples at module terminals.
J12 DS18B20 unchanged. PCB pending. NO POWER / NO FLASH / NO FIRE.

DRAFT: LM5013 5V is assigned to +5V_DEVKIT in this hierarchy. USB reverse-feed arbitration remains UNQUALIFIED. NO POWER.

TWO RELAYS: K1 GLOW; K2 CLUTCH. U11 IN1=GLOW_CMD, IN2=FUEL_CMD; IN3/IN4, ENBL and RESET tied to GND. No independent welded-contact cutoff. Motor control remains external HC160. DRAFT / NO POWER / NO FLASH / NO FIRE.

MAIN INPUT / ROOM SENSOR
J13: 1=VIN_RAW_24, 2=GND. Main design range 21.6..28.8V.
5V conversion is on-board; USB/external power arbitration remains unqualified.
J12: 1=3V3, 2=DQ, 3=GND. Never connect raw 24V to logic terminals.

RELAY DRIVER DRAFT / NO POWER / NO FLASH / NO FIRE
U11 DRV8803: GLOW_CMD to IN1; FUEL_CMD to IN2; IN3/IN4, nENBL and RESET to GND.
Exactly K1 glow and K2 clutch; no independent welded-contact cutoff.
Driver, relay-coil, transient and thermal qualification remain open.

REMOTE CONTROL / MAIN-POWER MONITOR – DRAFT
U20 monitors P24 presence; USB power paths remain unqualified.
External watchdog/latch and hardware ARM_PULSE logic have been removed.
Do not restore obsolete permission circuitry. No physical tests performed.

DEFINED INPUTS / OUTPUTS – REMOTE CONTROL ONLY
R1: room-sensor pull-up; R2/R3: thermocouple CS pull-ups.
R7/R8: fan/PWM pulldowns. Relay driver resistors remain inside MinimalSafety.
No display, encoder or Start/Stop button. Their six former DevKit pins are NC.
Remote commands are handled by ESP32; no new control connector is added.
Firmware and remote link-loss behavior remain to be implemented and tested.

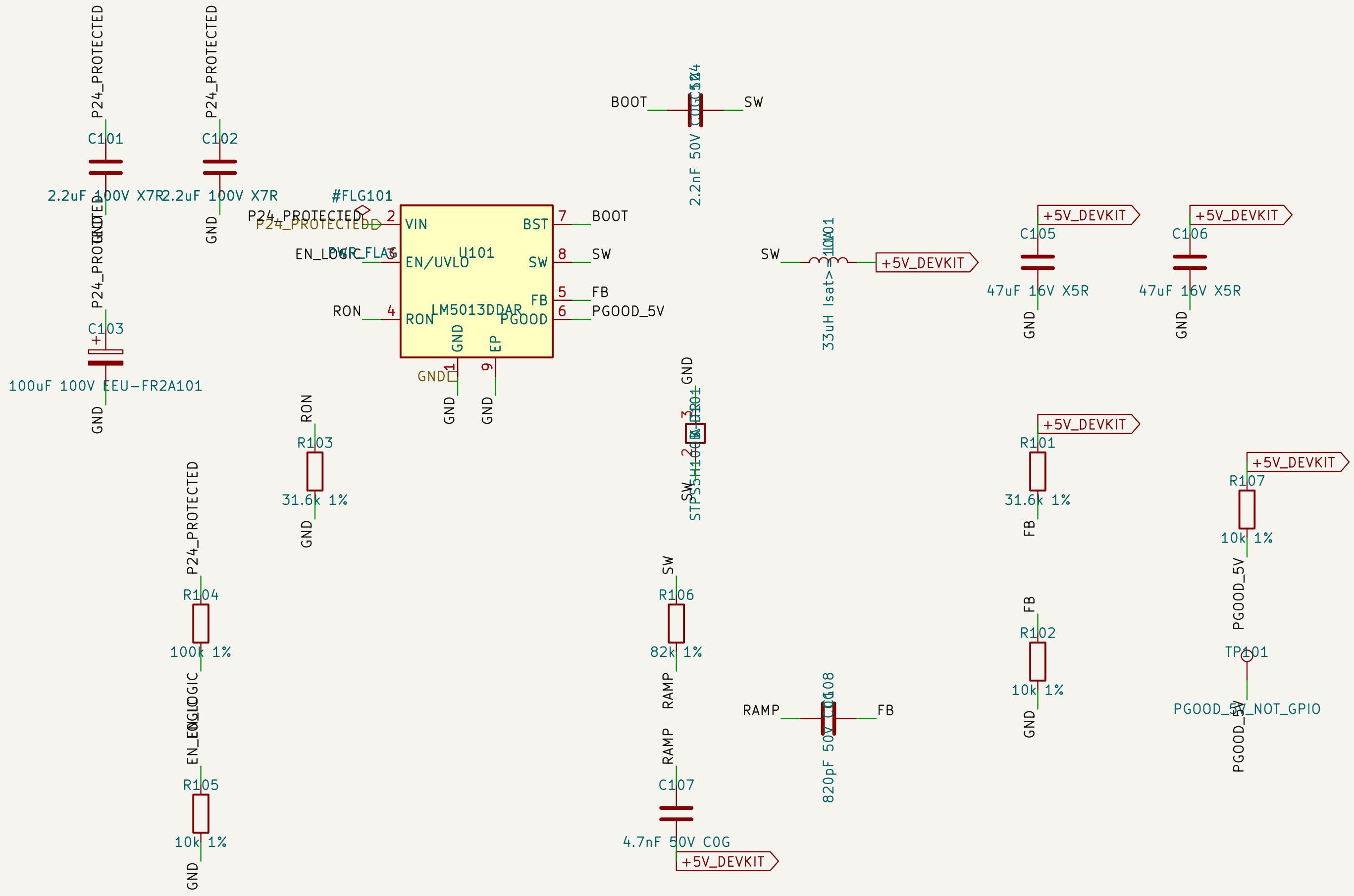
J27 GLOW: 1=switched positive; 2=power GND.
Schematic assigned; PCB current/transient qualification OPEN. NO POWER / NO FIRE.

J28 CLUTCH: 1=switched positive; 2=power GND.
Schematic assigned; PCB current/transient qualification OPEN. NO POWER / NO FIRE.

NO LOCAL USER CONTROLS / DRAFT
No display, Start/Stop, encoder or separate RESET on this carrier.
No replacement UI header, jumper or local arming control.
Old external watchdog/latch circuitry is not restored.
DevKit built-in EN/BOOT and USB programming circuitry are unchanged.
No automatic heater restart after reset or a restored remote link.

Interface circuit only; not a burner safety approval

Sheet: /
File: Sirokko26201_control_motor_candidate.kicad_sch
Title: Sirokko 262.01 – common controller interfaces
Size: A2
Date: 2026-10-23
Rev: DRAFT-01
KiCad E.D.A. 10.0.7
Id: 1/5

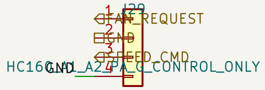

P24_PROTECTED
C101
2.2uF 100V X7R
GND
P24_PROTECTED
C102
2.2uF 100V X7R
GND
P24_PROTECTED
C103
100uF 100V EEU−FR2A101
GND
#FLG101
PWR_FLAG
P24_PROTECTED
EN_LOGIC
RON
U101
LM5013DDAR
VIN
EN/UVLO
RON
GND
EP
BST
SW
FB
PGOOD
2
3
4
1
9
7
8
5
6
BOOT
SW
FB
PGOOD_5V
GND
GND
GND
BOOT
SW
C104
2.2nF 50V C0G
SW
L101
33uH Isat>=1A
+5V_DEVKIT
C105
47uF 16V X5R
GND
+5V_DEVKIT
C106
47uF 16V X5R
GND
+5V_DEVKIT
D101
STPS3H100U
GND
SW
RON
R103
31.6k 1%
GND
P24_PROTECTED
R104
100k 1%
EN_LOGIC
R105
10k 1%
GND
SW
R106
82k 1%
RAMP
RAMP
C107
4.7nF 50V C0G
+5V_DEVKIT
RAMP
C108
820pF 50V C0G
FB
+5V_DEVKIT
R101
31.6k 1%
FB
FB
R102
10k 1%
GND
+5V_DEVKIT
R107
10k 1%
PGOOD_5V
TP101
PGOOD_5V_NOT_GPIO
PGOOD_5V
1
2
3
4
5
6
A
B
C
D

тание логики 24 В → 5 В / 1 А. DRAFT, НЕ ДЛЯ ПРОИЗВОДСТВА.
5013, SNVSCB3 (10.2022), §6-9.2.2.7. Расчётный вход 18-30 В.
о ПОСЛЕ входной защиты; выбросы должны быть ограничены до ≤50 В.
к DevKit и логике на общей PCB. Это не внешний вход 5 В.
онтрольная точка, НЕ GPIO ESP32; не заменяет контроль наличия 24 В.
отключение обратной подпитки USB и силовые каналы здесь ещё не реализованы.
ые места требуют окончательной квалификации; особенно L101 и D101.
щё не выполнена; это отдельный проектный лист, не отдельная внешняя плата.

Sheet: /LOGIC_POWER_24_TO_5V__DRAFT_NOT_FOR_ORDER/
File: logic_power_24_5_DRAFT.kicad_sch

Title:

Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 2/5


J29
1
FAN_REQUEST
2
GND
3
SPEED_CMD
4
HC160_A1_A2_PA_G_CONTROL_ONLY
GND

Sheet: /MOTOR_EXTERNAL_HC160/
File: motor_external_hc160.kicad_sch

Title:

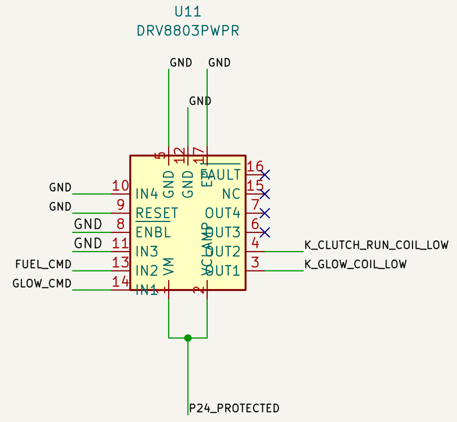
Size: A4
Date:
KiCad E.D.A. 10.0.7
Rev:
Id: 3/5

SIROKKO: MINIMAL SAFETY CORE – 3 IC + EXISTING RELAY DRIVER

DRAFT / NOT FOR ORDER / NO POWER / NO FIRE – hierarchy INTEGRATED in root SCH; PCB/firmware NOT synchronized
CURRENT CONTRACT: external TPS3431 watchdog, SN74HCS74 latch and SN74HCS11 safety logic REMOVED / CANCELLED.

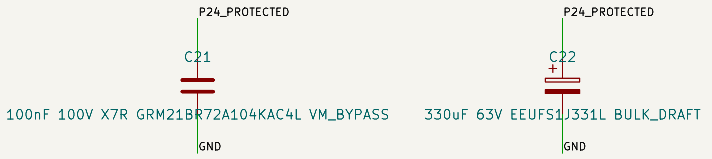
TWO–RELAY DRIVER: IN1=GLOW_CMD; IN2=FUEL_CMD; unused IN3/IN4 and ENBL/RESET=GND. R86/R87 pull commands LOW. OUT3/OUT4 unused. No independent welded–contact cutoff. DRAFT / NO POWER / NO FLASH / NO FIRE.

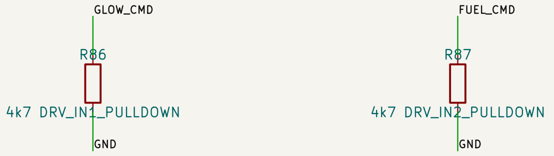
FUTURE FIRMWARE REQUIREMENT (NOT IMPLEMENTED HERE): ESP32–S3 IWDT+TWDT; supervise control task; panic/reboot; keep commands OFF and prohibit automatic restart after reset.

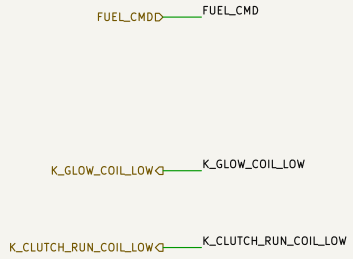

U11
DRV8803PWPR
GND
GND
GND
5
12
17
GND
GND
EP
16
FAULT
15
NC
7
OUT4
6
OUT3
4
OUT2
3
OUT1
K_CLUTCH_RUN_COIL_LOW
K_GLOW_COIL_LOW
GND
10
IN4
GND
9
RESET
GND
8
ENBL
GND
11
IN3
FUEL_CMD
13
IN2
GLOW_CMD
14
IN1
VM
VCLAMP
1
2
P24_PROTECTED

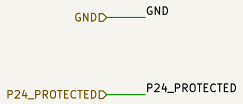
TWO–RELAY DRIVER: IN1=GLOW_CMD; IN2=FUEL_CMD; unused IN3/IN4 and ENBL/RESET=GND. R86/R87 pull commands LOW. OUT3/OUT4 unused. No independent welded–contact cutoff. DRAFT / NO POWER / NO FLASH / NO FIRE.

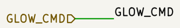
Process checks, timings, purge and software fault handling belong to ESP32; motor/purge power path is unchanged.


P24_PROTECTED
C21
100nF 100V X7R GRM21BR72A104KAC4L VM_BYPASS
GND
P24_PROTECTED
C22
+
330uF 63V EEUFS1J331L BULK_DRAFT
GND


GLOW_CMD
R86
4k7 DRV_IN1_PULLDOWN
GND
FUEL_CMD
R87
4k7 DRV_IN2_PULLDOWN
GND


FUEL_CMD
FUEL_CMD
K_GLOW_COIL_LOW
K_GLOW_COIL_LOW
K_CLUTCH_RUN_COIL_LOW
K_CLUTCH_RUN_COIL_LOW


GND
GND
P24_PROTECTED
P24_PROTECTED


GLOW_CMD
GLOW_CMD

TWO–RELAY DRIVER: IN1=GLOW_CMD; IN2=FUEL_CMD; unused IN3/IN4 and ENBL/RESET=GND. R86/R87 pull commands LOW. OUT3/OUT4 unused. No independent welded–contact cutoff. DRAFT / NO POWER / NO FLASH / NO FIRE.

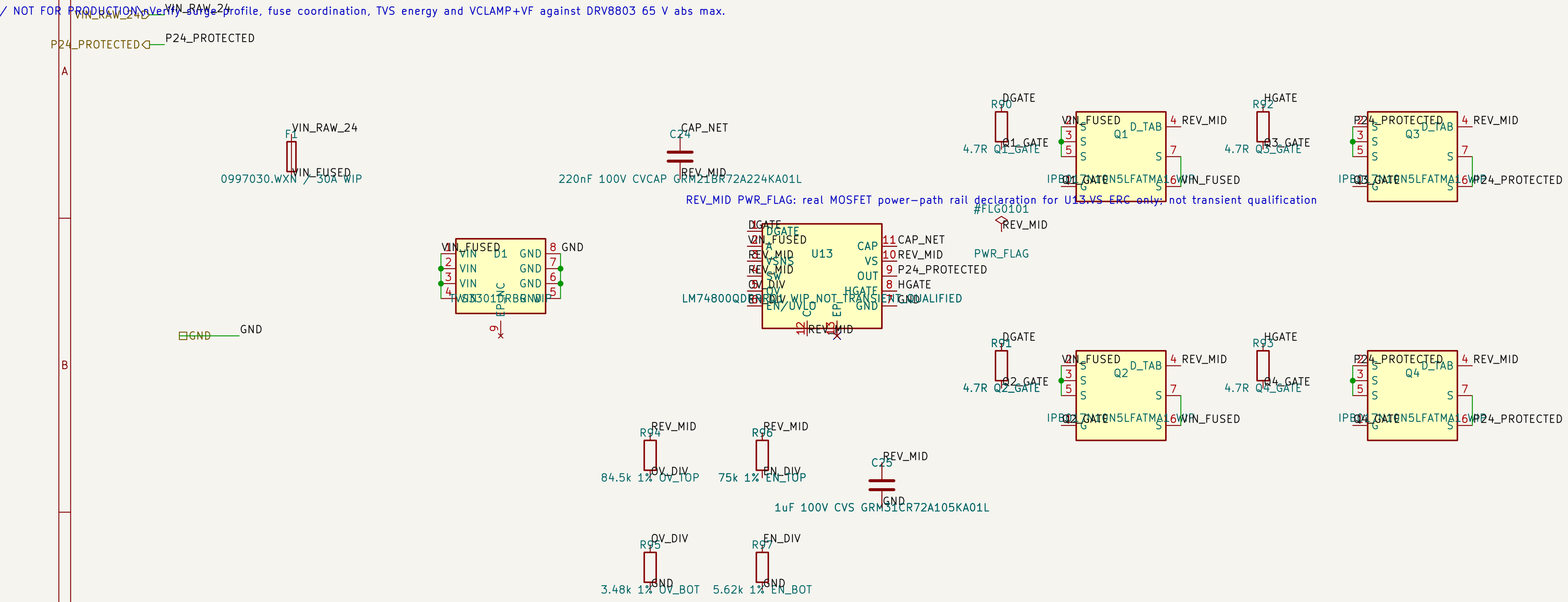
24V loss/recovery, startup dynamics, relay ratings and full safety acceptance remain OPEN.

Sheet: /MinimalSafety/
File: MinimalSafety.kicad_sch

Title:

Size: A3 Date:
KiCad E.D.A. 10.0.7

Rev:
Id: 4/5


/ NOT FOR PRODUCTION\nVerify surge profile, fuse coordination, TVS energy and VCLAMP+VF against DRV8803 65 V abs max.
VIN_RAW_24
VIN_RAW_24
P24_PROTECTED
P24_PROTECTED
F1
VIN_RAW_24
VIN_FUSED
0997030.WXN / 30A WIP
C24
CAP_NET
REV_MID
220nF 100V CVCAP GRM21BR72A224KA01L
REV_MID PWR_FLAG: real MOSFET power-path rail declaration for U13.VS ERC only; not transient qualification
#FLG0101
REV_MID
PWR_FLAG
D1
VIN_FUSED
VIN
VIN
VIN
GND
GND
GND
8 GND
TVS3301DRBR WIP
EP/NC
9
U13
DGATE
VIN_FUSED
REV_MID
REV_MID
OV_DIV
EN_DIV
A
VSNS
SW
OV
EN/UVLO
C
EP
CAP
VS
OUT
HGATE
GND
11 CAP_NET
10 REV_MID
9 P24_PROTECTED
8 HGATE
7 GND
12
13
REV_MID
LM74800QDRRRQ1 WIP NOT TRANSIENT QUALIFIED
GND
GND
R90
DGATE
Q1_GATE
4.7R Q1_GATE
Q1
VIN_FUSED
D_TAB
4 REV_MID
S
G
7
6
VIN_FUSED
IPB017N10N5LFATMA1 WIP
Q1_GATE
R92
HGATE
Q3_GATE
4.7R Q3_GATE
Q3
P24_PROTECTED
D_TAB
4 REV_MID
P24_PROTECTED
IPB017N10N5LFATMA1 WIP
Q3_GATE
R91
DGATE
Q2_GATE
4.7R Q2_GATE
Q2
VIN_FUSED
D_TAB
4 REV_MID
VIN_FUSED
IPB017N10N5LFATMA1 WIP
Q2_GATE
R93
HGATE
Q4_GATE
4.7R Q4_GATE
Q4
P24_PROTECTED
D_TAB
4 REV_MID
P24_PROTECTED
IPB017N10N5LFATMA1 WIP
Q4_GATE
R94
REV_MID
OV_DIV
84.5k 1% OV_TOP
R96
REV_MID
EN_DIV
75k 1% EN_TOP
C25
REV_MID
GND
1uF 100V CVS GRM31CR72A105KA01L
R95
OV_DIV
GND
3.48k 1% OV_BOT
R97
EN_DIV
GND
5.62k 1% EN_BOT

Sheet: /INPUT_PROTECTION_WIP_NOT_QUALIFIED/
File: Sirokko_Input_Protection_WIP.kicad_sch

Title:

Size: A4 Date:
KiCad E.D.A. 10.0.7

Rev:
Id: 7/5